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Diseases Burden and epidemiological transition status at the national and sub-national level in India: a contemporary perspective

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Abstract
The present study aims to assess age and gender-specific disease burden variations in India in 2020 and evaluate the epidemiological transition status nationwide and across selected states. Utilising data from vital statistics, census, and cause of death reports and applying innovative methods, the present study estimated cause-specific mortality patterns, Years of Life Lost (YLL), Years Lived with Disability (YLD), and Disability-adjusted Life Years (DALYs). The study categorised over two hundred diseases into five groups: Infectious, Nutritional, Maternal, and Perinatal Diseases (INMPDs), NCDs, Injuries, Ill-defined, and COVID-19. Further, to determine the epidemiological transition status, the Epidemiological Transition Ratio (ETR) calculates the ratio of DALYs from INMPDs to NCDs. A lower ETR value indicates a higher epidemiological transition state and vice-versa. Among all causes, NCDs were predominant, causing 60% of deaths, YLLs (51%, 98.7 million), YLDs (71%, 79 million), and DALYs (58%, 177 million). INMPDs, Injuries, COVID-19, and other diseases contributed 5%-20% to these measures. This trend was consistent at the sub-national level. Age-wise DALY distribution showed an inverse relationship between INMPDs and NCDs, with younger populations being more affected by INMPDs and older populations being affected by NCDs. Gender analysis revealed higher YLLs and DALYs in males, while females had higher YLDs. Leading causes included CVDs, maternal-neonatal disorders, chronic respiratory diseases, and infectious diseases, with top individual causes being diabetes, COVID-19, Ischemic-heart Disease, Hypertension, and Tuberculosis. The ETR revealed 36 DALYs of INMPD per 100 DALYs of NCDs. It revealed that India is in a high-middle epidemiological transition state with a growing NCD burden. ETR varied eloquently by state, from Kerala's lowest ratio (7) to Uttar Pradesh's highest (90). The present study highlights a significant burden towards chronic diseases, with variations across states and pronounced age and gender disparities. The epidemiological transition status across Indian states spans from developing to developed, with at least a decade of lag between them, emphasising the need for tailored health strategies. Enhancing the health system with a focus on prevention and early detection of chronic illnesses is essential, without neglecting communicable diseases, to improve health outcomes, particularly in underserved regions and vulnerable populations. Our findings underscore the importance of region-specific policies and interventions to effectively address the diverse health challenges across India.

Keywords Diseases Burden · Epidemiological transition · Mortality · NCDs · DALYs · India

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1 Introduction

Globally, The Burden of Diseases (BOD) is the health problem measured by the mortality, morbidity, disability, and financial cost behind all health problems and other indicators. It is typically assessed using incidence, prevalence, and Disability-Adjusted Life Years (DALYs). Such indicators and work need to be conducted periodically. They aid in planning preventive actions, comparing different health interventions and research, assessing the status of healthcare systems, identifying high-risk populations, and planning for future health needs [1, 2]. Additionally, monitoring the shift in disease burden helps assess epidemiological transitions across different regions and over time [3].

More precisely, the epidemiological transition focuses on the complex change in patterns and trends of health and illness. However, various stages of epidemiological transition can easily be understood by shifting the burden of disease patterns from Communicable Diseases (CDs) to Non-Communicable Diseases (NCDs) [4, 5]. High-income countries of the developed world have successfully undergone the epidemiological transition from infectious diseases to chronic degenerative diseases. However, many developing nations have not yet achieved this transition. Improved hygiene practices, modern sanitation, antibiotics, effective vaccination, balanced diets, and public healthcare have considerably reduced the burden of infectious diseases in developed countries [6]. Despite these advancements, the emergence of HIV, haemorrhagic fevers, Ebola, Severe Acute Respiratory Syndrome (SARS), COVID-19, avian influenza, and other diseases. It has increased the burden of infectious diseases globally, affecting both developing and developed nations [7–9].

Among the various models, Omran has suggested a delayed variant transition in the epidemiological model specifically for developing South Asian countries like India, Pakistan, and others [10]. India's morbidity patterns have shifted significantly since independence. Until the 1980s, infectious, maternal, perinatal, and parasitic diseases dominated. However, by 1990, NCDs overtook communicable diseases. Visaria's study highlights this dual burden, showing a decline in mortality from infectious diseases (48% in 1969 to 22% in 1995) alongside a rise in NCDs (36%–55%) [11]. According to a BOD study, during 1990–2016, the NCD burden rose while CDs declined in most Indian states. Some states like Madhya Pradesh, Bihar, Uttar Pradesh, and Rajasthan face a dual burden. Initially, all states had a higher burden of CDs, but by 2016, NCDs and injuries outweighed CMNNDs [3]. Another study revealed that India experienced approximately 486 million DALYs and 9.7 million deaths in 2017. Of total national DALYs, Communicable, nutritional maternal, and perinatal causes accounted for 35%, while NCDs comprised 47% and injuries comprised 11% of DALYs overall [12]. Also, the study found eloquent variation at the state level in disease burden.

India is the world's most populous and heterogeneous country, with over 1.42 billion in population [13]. India's diverse states show significant socio-economic, demographic, dietary, hygiene, and ecological variations impacting health outcomes. Timely measurement is crucial for effective planning, but poor-quality vital statistics hinder epidemiological studies and the understanding of shifts in morbidity and mortality [14, 16]. While numerous studies address India's disease burden and epidemiological transition nationally, few peer-reviewed works focus on the state level [3, 12], precisely age and gender perspectives. Still, an ample understanding of changes in death patterns, disease burden, and cause-specific mortality with robust, large-scale data for each state of India is not readily available. In India, mortality data, especially the causes of death, is lacking and incomplete, even in the civil registration record [17, 18]. Further, research is required to understand these measures using limited quality of data and innovative indirect methods. Thus, the primary objective of our study is to assess the age and gender-specific variations in the disease burden in 2020 for India and selected states. Secondly, the study evaluated the epidemiological transition status in India in the contemporary period.

2 Data and methodology

We have gathered data from many sources to thoroughly grasp the study's objectives. Our compilation includes datasets and reports such as the World Population Prospect (WPP) 2022, Technical Census Population Projection (CPP) spanning 2011–2036, the Sample Registration System (SRS) Statistical Report, and the Medical Certification of Cause of Death (MCCD) for 2020. Additionally, we've incorporated the SRS abridged life table spanning 2016–2020 for India's entire population and states. It's important to note that all these datasets originate from authenticated government sources, providing comprehensive demographic and mortality insights at both national and sub-national levels. Their validity, accuracy and completeness are publicly available. For further elucidation on these datasets, detailed information is available within the respective reports [17, 19–22].

The study explored the disease burden and the current state of the epidemiological transition. It estimated various parameters such as population count, the number of deaths, Causes of Death (COD), Years of Life Lost (YLL) due to premature mortality, Years Lived with Disabilities (YLD), and Disability-Adjusted Life Years (DALYs). These metrics are calculated for both national and sub-national levels and also for the leading causes of death in India, considering different age groups and sex in the Indian population for the year 2020. The study focuses on the twenty bigger states at the state level due to data unavailability and incompleteness for all Indian states. Specifically, states with better death data availability and MCCD reporting of at least 5% were included in the analysis. Furthermore, the study elucidates the step-by-step procedure for estimating DALYs. To achieve this goal, the study first estimates other essential indicators, including population figures, the number of deaths, and the causes of death in India and the selected states.

Population Estimation at the India and state level: To address the underestimation in the Census Population Projection (CPP) for the current period, we estimated the population for 2020. We obtained the latest population figures for India from the World Population Projection (WPP) and utilised state population shares and age-sex distributions from the CPP to estimate the population of individual states, aligning them with India's overall population from the WPP.
Deaths Estimation at the India and state level: To estimate age-sex-wise deaths for the entire population in 2020. We utilised the Age-Specific Death Rates (ASDR) for both states and India, compiled from the Sample Registration System (SRS) statistical report. These rates were then multiplied by the population estimates obtained in the previous step, enabling us to calculate the number of deaths across different age and sex categories.
Diseases Classification and Cause of Death Distribution: We turned to the MCCD report for 2020 to obtain the proportion of age-specific deaths. This report contains detailed information on over 220 diseases, covering nearly all causes of death in India. For our study's purposes, we classified these diseases into five main groups based on the International Classification of Diseases (ICD-10) and the Indian Council of Medical Research (ICMR) technical group classification [12]. The first group encompasses Infectious, Nutritional, Maternal, and Perinatal diseases (INMPDs), the second group consists of Non-Communicable Diseases (NCDs), and the third group comprises Injuries. Additionally, we retained a fourth group for Ill-defined causes of death, and the fifth group specifically focused on COVID-19 and other emerging diseases.
We calculated age-sex-cause-specific death proportions for India in 2020 to analyse the burden of disease comprehensively. At the state level, the MCCD provided only sex-wise cause-of-death data without age group details. To address this, we indirectly estimated age-sex death proportions by aligning each disease's state-level deaths with its major disease group distribution. After splitting deaths into age-sex groups, we determined state-wide deaths for each disease. The classification and estimation procedure followed the same approach as for the national-level analysis, resulting in age-sex-cause-specific death proportions for all selected states in 2020
Number of Cause of Deaths in Population: To determine the actual number of age-sex-cause-specific deaths within the population, we multiplied the estimated age-sex-wise deaths for the entire population by the age-sex-cause-specific death proportions corresponding to the different major causes of death for 2020.
Estimation of YLL: To calculate YLL, we used estimates of age-sex-cause-specific deaths for India and selected states. We also gathered age and sex-specific Life Expectancy (LE) data from the SRS abridged life table. Finally, we multiplied these indicators to obtain the YLLs for India and the selected states.
Estimation of YLD: We employed the indirect innovative estimation method due to a lack of data on disability weights, incidences, etc. Previously, the ICMR technical advisory committee had referred to the indirect estimation of YLD [12]. For INMPDs, NCDs, and Injuries, we mapped and extracted the (YLD/YLL) ratio for India from the GBD study conducted in 2017 [12]. We utilised another study for COVID-19 and other disease ratios [23, 24]. The ratios were extracted according to the age-sex and major group distribution. For the state level, we assumed that India's (YLD/YLL) ratio remains constant across all states. Finally, to obtain YLDs for both India and selected states, we multiplied our estimated Years of Life Lost (YLL) by the (YLD/YLL) ratio.
Estimation of DALY: In the last step of the DALY calculation, we have added YLL and YLD according to age and sex for India and selected states.

The study used YLLs, YLDs, and DALYs to analyse the age-sex-wise disease burden across major groups: INMPDs, NCDs, and Injuries. The fourth group included Ill-defined causes, and a fifth included COVID-19 and emerging diseases. Leading causes of death were estimated at both group and individual levels. The study's ultimate objective was to propose and assess the Epidemiological Transition Ratio (ETR) nationally and sub-nationally, calculated as the ratio of DALYs from INMPDs to NCDs. An inverse relationship indicates a lower ratio, which signifies a higher epidemiological transition

status. This operational rationality was supported by another study [3]. This novel indirect method and ETR ratio help to understand the epidemiological health status. The lower the ETR value reflects, the higher the burden of NCD compared to the INMPDs and vice-versa, i.e., advanced epidemiolocal stage. These methods were adopted to address given data limitations effectively. Moreover, the National Disease Burden Initiative validated and endorsed our methodology [12].

3 Results

Table 1 represents the distribution of causes of death at both the national and state level in India. The data reveals significant differences among the major causes of death across the country and within individual states. Nationally, NCDs account for the highest proportion of deaths at approximately 60%. Notably, there is a gender differential in NCD-related deaths, with women accounting for a higher share (62.6%) compared to men (58.8%). Following NCDs, the leading causes of death were INMPDs, which account for 15%, Injuries at 9%, COVID-19 and other new diseases at 8.5%; and lastly, ill-defined and other diseases at 7%. At the state level, a similar pattern is observed, with NCDs accounting for a prominent share of deaths, ranging from 40% in Madhya Pradesh to 75% in Kerala. However, there is a reverse range in the case of INMPDs, which range from 6% in Kerala to 23% in Madhya Pradesh.

Table 2 presents age-wise statistics for all indicators, revealing a slightly bimodal disease burden distribution. This burden is higher in the early ages of life (under one year), declines as age increases, and upsurges again from those aged 45 years and above in India. The present study estimated that around 8.4 million deaths occurred in India in 2020, with 4.75 million males and 3.65 million females. Further, the estimated disease burden observed was around 195 million YLLs, 111 million YLDs, and 306 million DALYs, with males accounting for 157 million and females for 149 million.

Tables 3, 4, and 5 focused on the estimates of absolute YLLs, YLDs, and DALYs distribution by major causes and gender across India and selected states, respectively. NCDs have the highest burden, with 98.7 million YLLs, 79 million YLDs, and 177 million DALYs. NCDs are followed by INMPDs with 63 million DALYs, injuries with 30 million, ill-defined and other diseases with 22.7 million, and COVID-19 and other emerging diseases with 12 million DALYs. Among all states, Uttar Pradesh accounted for the highest absolute burden and gender differential for all kinds of diseases, especially dominant figures of NCDs are YLLs (11.7 million), YLDs (10.3 million) and DALYs (22 million), followed by similar statistics in Maharashtra, Madhya Pradesh, Tamil Nadu and other states. The gender differential was addressed using data from Tables 3 and 5, where YLLs and DALYs are higher for males across all major causes (INMPDs, NCDs, Injuries, Ill-defined others and Covid-19 and other new diseases) compared to females. However, Table 4 shows that YLDs are higher for females than males, except for COVID-19 and other new diseases.

Furthermore, Table 6 provides the distribution of DALYs by age and gender according to the major causes in India. Overall, the DALYs share of INMPDs account for 21% of the burden, NCDs for 58%, Injuries for 10%, Ill-defined and Other causes for 4%, and COVID-19 and other emerging diseases for 4%. It shows that the burdens of NCDs are much higher and more than double that of INMPDs and more than five times that of injuries in India in contemporary times. Age-wise, DALYs for INMPDs are significantly higher in infants (under one year) compared to other age groups. Conversely, NCDs and Injuries have lower DALYs at early ages but increase with age, becoming most prevalent in individuals aged 45–54 years and older. Gender-specific, DALYs in millions are estimated as INMPDs (male: 32; female: 31.3), NCDs (male: 89.5; female: 88.3), injuries (male: 15.5; female: 15), ill-defined and other causes (male: 12.2; female: 10.5), and COVID-19 and other new diseases (male: 7.7; female: 4.5). This distribution reflects that the burden of diseases is considerably higher among males compared to females across all major causes.

Table 7 reveals the leading causes of burden in India, both for the group and individual causes. Among the top group causes, the leading cause of burden is Cardiovascular Diseases (CVDs), accounting for over 50 million DALYs. Followed by Maternal and Neonatal Conditions with 32 million DALYs. Chronic Respiratory Diseases (CRD) contribute to 28.8 million DALYs. Ranked by DALYs, the following causes are Infectious Diseases (16.5 million), Neurological Disorders (14.6 million), Cancers and Other Neoplasms (7.4 million), and Digestive Diseases (6.7 million). At the individual causes, the top five causes ranked by DALYs are Diabetes Mellitus (14.7 million), COVID-19 (10.6 million), Ischemic Heart Disease (10.5 million), Hypertensive Heart Disease (5.65 million), and Tuberculosis (3.3 million).

Figure 1 represents the epidemiological transition status through the Epidemiological Transition Ratio (ETR) at national and sub-national levels in India. The ETR value for India is around 36, similar for both males and females, indicating 36 DALYs of INMPDs per 100 DALYs of NCDs. At the state level, the highest ETR is found in Uttar Pradesh at 90 (male: 79 and female: 101), reflecting a significant dual burden of INMPDs and NCDs, especially among females. It followed by Madhya Pradesh with an ETR of 69 (male: 71 and female: 67), Rajasthan also at 69 (male: 64 and female: 74), and Assam

Table 1 Share of cause of death distribution by gender and major group causes in India and selected states, 2020

Major group	INMPDs (%)			NCDs (%)			Injuries (%)			Ill-defined and other (%)			Covid-19 and other new (%)		
Country/state	Male	Female	Person	Male	Female	Person	Male	Female	Person	Male	Female	Person	Male	Female	Person
India	14.0	16.7	15.0	58.8	62.6	60.1	10.4	6.5	9.0	7.4	7.2	7.3	9.4	6.9	8.5
andhra pradesh	10.6	13.6	11.7	70.4	71.6	70.8	5.2	3.5	4.6	1.1	1.3	1.1	12.7	10.0	11.7
Assam	14.9	13.1	14.3	69.3	71.2	69.9	13.6	12.6	13.3	0.1	0.2	0.1	2.1	3.0	2.4
Chhattisgarh	17.2	22.7	19.2	47.7	47.7	47.7	6.6	5.0	6.0	21.7	19.9	21.0	6.8	4.7	6.0
Delhi	19.1	22.1	20.2	49.1	50.8	49.7	9.9	8.5	9.4	11.4	9.4	10.6	10.6	9.3	10.1
Gujarat	21.5	22.3	21.8	60.4	61.4	60.7	3.5	2.6	3.2	10.2	10.6	10.3	4.4	3.2	3.9
Haryana	12.4	14.7	13.2	45.2	50.8	47.0	10.6	5.5	8.9	28.1	25.1	27.1	3.8	3.8	3.8
Himachal Pradesh	14.9	14.9	14.9	55.3	60.6	57.3	6.3	4.3	5.5	8.9	9.6	9.1	14.7	10.6	13.1
Jharkhand	9.2	9.1	9.2	58.9	63.1	60.5	24.0	23.6	23.8	0.7	0.5	0.7	7.1	3.7	5.9
Karnataka	15.6	17.8	16.4	62.3	65.7	63.5	9.8	6.8	8.8	1.9	2.4	2.1	10.4	7.3	9.3
Kerala	5.9	6.4	6.1	74.0	78.2	75.6	7.7	5.6	6.9	4.5	3.2	4.0	7.8	6.6	7.3
Madhya Pradesh	21.1	24.8	22.3	41.0	39.2	40.4	6.9	5.8	6.5	28.4	28.6	28.5	2.6	1.7	2.3
Maharashtra	9.6	11.7	10.3	56.2	67.2	59.6	10.6	1.8	7.8	5.5	4.8	5.3	18.1	14.5	17.0
Odisha	23.5	26.2	24.4	57.8	56.3	57.3	5.4	5.8	5.5	10.5	10.2	10.4	2.9	1.6	2.4
Punjab	18.5	18.5	18.5	57.1	62.4	59.2	7.3	6.2	6.8	4.4	3.5	4.0	12.8	9.5	11.4
Rajasthan	29.2	36.7	31.9	45.6	42.2	44.4	11.4	7.3	9.9	7.8	9.2	8.3	5.9	4.6	5.5
Tamil Nadu	8.4	7.6	8.1	73.6	77.5	75.1	5.7	3.5	4.9	9.0	9.4	9.1	3.3	1.8	2.8
Telangana	8.1	10.3	8.9	60.4	61.6	60.8	18.8	17.9	18.5	4.8	3.8	4.4	7.9	6.4	7.4
Uttarakhand	18.9	21.8	20.1	59.5	60.6	59.9	3.5	3.3	3.4	3.8	4.3	4.0	14.3	10.0	12.6
Uttar Pradesh	12.8	24.0	16.7	30.8	33.8	31.8	37.6	25.1	33.2	3.9	3.4	3.7	14.9	13.7	14.5
West Bengal	21.1	26.2	23.1	58.7	56.7	57.9	5.6	5.8	5.7	9.8	8.2	9.2	4.8	3.1	4.1

INMPD Inflectional, Nutritional, Maternal, Perinatal Diseases; *NCDs* Non-Communicable Diseases

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Table 2 Estimated deaths, YLL, YLDs and DALYs distribution by gender and age group wise in India, 2020

BOD Measures	Sex	< 1	1–4	5–14	15–24	25–34	35–44	45–54	55–64	65–69	70 +	All ages
Deaths	Male	324,222	37,680	38,965	113,840	185,531	306,599	500,614	854,338	546,221	1,850,572	4,758,583
	Female	285,723	39,436	37,762	86,501	110,685	153,834	273,586	569,076	393,891	1,696,611	3,647,104
	Persons	609,945	77,116	76,727	200,342	296,217	460,433	774,200	1,423,414	940,112	3,547,182	8,405,687
YLLs	Male	22,242,468	2,638,809	2,456,459	6,076,760	8,157,115	10,680,196	13,119,815	15,728,709	7,756,335	20,132,642	108,989,308
	Female	20,413,138	2,883,463	2,504,758	4,904,265	5,224,305	5,813,877	7,845,976	11,602,794	6,105,309	18,978,505	86,276,390
	Persons	42,655,607	5,522,273	4,961,217	10,981,025	13,381,420	16,494,073	20,965,791	27,331,502	13,861,644	39,111,147	195,265,698
YLDs	Male	915,456	2,063,865	4,651,858	6,482,815	6,633,052	6,796,315	6,291,232	5,644,688	2,466,086	6,083,186	48,028,555
	Female	951,213	2,465,529	5,962,108	10,547,537	11,396,757	8,696,736	6,809,673	6,660,873	2,815,611	7,074,671	63,380,707
	Persons	1,866,669	4,529,394	10,613,966	17,030,352	18,029,809	15,493,051	13,100,905	12,305,561	5,281,696	13,157,858	111,409,262
DALYs	Male	23,157,924	4,702,675	7,108,317	12,559,575	14,790,167	17,476,511	19,411,047	21,373,397	10,222,421	26,215,828	157,017,863
	Female	21,364,352	5,348,992	8,466,866	15,451,802	16,621,062	14,510,613	14,655,648	18,263,666	8,920,920	26,053,176	149,657,097
	Persons	44,522,276	10,051,667	15,575,183	28,011,377	31,411,229	31,987,124	34,066,696	39,637,063	19,143,341	52,269,004	306,674,960

YLL Years of Life Lost due to Premature Mortality; *YLD* Years of Healthy Life Lost due to Disability; *DALYs* Disability Adjusted Life Years

Table 3 Estimated YLLs distribution by gender and major group causes in India and selected states, 2020

Disease Categories	INMPDs			NCDs			Injuries			Ill-defined other			Covid-19 and other new		
Gender	Male	Female	Person	Male	Female	Person	Male	Female	Person	Male	Female	Person	Male	Female	Person
India	27,665,998	25,729,670	53,395,668	54,252,194	44,445,211	98,697,404	11,379,448	5,507,415	16,886,863	7,974,835	6,116,323	14,091,157	7,716,834	4,477,771	12,194,606
Andhra Pradesh	807,201	598,465	1,405,666	2,609,824	1,999,417	4,609,241	232,937	100,689	333,626	52,517	43,081	95,598	440,786	271,194	711,979
Assam	700,120	793,467	1,493,588	1,671,339	789,720	2,461,059	470,568	929,073	1,399,641	17,601	17,724	35,325	51,442	130,731	182,172
Chhattis-garh	594,818	654,546	1,249,364	1,165,946	1,077,889	2,243,835	164,378	110,897	275,276	550,890	493,086	1,043,977	158,951	99,499	258,450
Delhi	188,913	175,144	364,057	458,690	374,923	833,612	80,239	58,579	138,818	105,862	70,418	176,280	105,189	68,214	173,402
Gujarat	1,687,636	1,267,539	2,955,175	2,809,292	2,100,004	4,909,296	215,532	100,409	315,942	454,475	347,368	801,843	178,908	100,550	279,458
Haryana	403,676	309,799	713,475	1,142,155	830,186	1,972,341	288,710	82,383	371,093	734,112	407,681	1,141,793	76,829	48,378	125,207
Himachal Pradesh	117,844	67,153	184,997	327,973	274,637	602,610	42,356	20,738	63,095	63,824	43,447	107,271	81,414	46,935	128,349
Jharkhand	596,408	456,619	1,053,027	1,194,027	1,220,817	2,414,844	665,251	686,989	1,352,240	18,266	14,068	32,334	126,767	61,945	188,712
Karnataka	985,876	820,070	1,805,946	2,841,322	2,047,342	4,888,664	581,355	234,771	816,126	87,690	80,059	167,749	431,800	213,141	644,941
Kerala	211,285	110,010	321,296	1,837,066	1,525,018	3,362,084	256,001	133,141	389,142	127,207	68,230	195,437	182,079	126,643	308,722
Madhya Pradesh	3,160,642	2,758,728	5,919,370	2,667,743	2,010,158	4,677,901	492,893	321,738	814,630	1,905,226	1,504,113	3,409,339	160,227	78,558	238,785
Maharash-tra	1,236,217	1,072,611	2,308,828	4,429,017	3,513,877	7,942,894	1,129,286	137,874	1,267,159	492,931	283,540	776,471	1,349,702	737,638	2,087,339
Odisha	1,373,127	1,192,815	2,565,942	2,403,217	2,036,724	4,439,941	242,213	192,354	434,567	454,492	356,848	811,341	114,587	55,389	169,977
Punjab	601,581	475,618	1,077,199	1,332,986	911,736	2,244,722	220,376	121,981	342,357	114,174	56,439	170,614	288,910	133,917	422,827
Rajasthan	2,029,999	2,295,408	4,325,407	2,133,216	1,725,362	3,858,578	623,237	344,132	967,369	390,234	400,453	790,688	240,351	154,772	395,122
Tamil Nadu	720,381	526,886	1,247,267	3,335,574	2,276,051	5,611,625	388,810	161,390	550,200	365,367	255,200	620,568	142,258	56,083	198,341
Telangana	479,328	354,338	833,667	1,192,450	820,601	2,013,051	471,776	290,580	762,356	92,852	50,134	142,986	153,107	84,688	237,795
Uttara-khand	202,524	188,202	390,726	496,933	386,304	883,237	41,720	24,518	66,237	35,479	29,895	65,374	115,535	66,328	181,863
Uttar PRADESH	7,627,882	9,044,773	16,672,655	6,409,873	5,340,866	11,750,739	3,231,548	2,452,556	5,684,105	709,225	623,878	1,333,102	2,878,329	1,943,590	4,821,919
West Bengal	1,539,797	1,271,703	2,811,500	3,665,274	2,698,852	6,364,127	378,242	253,911	632,153	657,193	372,321	1,029,514	286,762	150,632	437,394

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Table 4 Estimated YLDs distribution by gender and major group causes in India and selected major states, 2020

Major causes	INMPDs			NCDs			Injuries			Ill-defined other			Covid-19 and other new		
gender	Male	Female	Person	Male	Female	Person	Male	Female	Person	Male	Female	Person	Male	Female	Person
India	4,354,904	5,613,553	9,968,457	35,268,242	43,754,627	79,022,869	4,119,914	9,514,270	13,634,185	4,226,662	4,464,915	8,691,578	58,833	33,341	92,174
Andhra Pradesh	120,131	151,568	271,699	1,799,370	1,869,680	3,669,050	77,973	144,825	222,799	27,834	31,449	59,283	3361	2019	5380
Assam	149,280	210,685	359,965	890,772	650,046	1,540,818	146,369	1,644,452	1,790,821	9328	12,939	22,267	392	973	1366
Chhattisgarh	92,235	135,795	228,030	855,168	1,279,492	2,134,661	65,224	172,472	237,696	291,972	359,953	651,925	1212	741	1953
Delhi	44,890	66,270	111,160	246,696	378,043	624,739	34,278	83,868	118,146	56,107	51,405	107,512	802	508	1310
Gujarat	289,428	280,209	569,636	1,962,605	2,026,698	3,989,303	69,278	196,918	266,196	240,872	253,578	494,450	1364	749	2113
Haryana	87,532	81,304	168,835	700,486	696,578	1,397,064	96,012	137,734	233,746	389,079	297,607	686,686	586	360	946
Himachal Pradesh	19,976	21,484	41,461	198,979	223,070	422,049	14,937	28,352	43,289	33,827	31,716	65,543	621	349	970
Jharkhand	88,008	111,945	199,953	676,079	1,245,558	1,921,637	203,552	1,137,843	1,341,396	9681	10,269	19,950	966	461	1428
Karnataka	217,201	201,781	418,982	2,045,598	1,893,190	3,938,788	191,346	420,591	611,937	46,476	58,443	104,918	3292	1587	4879
Kerala	39,473	41,280	80,753	1,071,881	1,238,525	2,310,406	89,714	251,569	341,283	67,420	49,808	117,228	1388	943	2331
Madhya Pradesh	297,767	341,521	639,289	2,187,139	2,595,014	4,782,153	165,184	554,501	719,685	1,009,770	1,098,002	2,107,772	1222	585	1806
Maharashtra	269,543	353,695	623,238	2,719,658	3,150,727	5,870,385	376,348	270,953	647,301	261,253	206,984	468,238	10,290	5492	15,782
Odisha	199,922	238,355	438,277	1,736,801	2,213,083	3,949,884	89,421	358,982	448,403	240,881	260,499	501,380	874	412	1286
Punjab	156,980	147,800	304,780	849,362	806,973	1,656,335	79,686	192,076	271,761	60,512	41,201	101,713	2203	997	3200
Rajasthan	309,598	403,157	712,755	1,533,005	1,927,539	3,460,544	196,139	633,382	829,521	206,824	292,331	499,155	1832	1152	2985
Tamil Nadu	133,368	116,186	249,554	2,115,576	2,171,490	4,287,066	135,484	339,180	474,664	193,645	186,296	379,941	1085	418	1502
Telangana	51,112	59,406	110,518	897,035	878,718	1,775,753	170,878	427,929	598,807	49,212	36,598	85,809	1167	631	1798
Uttarakhand	49,926	60,635	110,561	333,119	385,694	718,813	13,009	53,623	66,632	18,804	21,823	40,627	881	494	1375
Uttar Pradesh	1,001,821	2,134,713	3,136,534	4,540,093	5,730,458	10,270,551	1,718,860	1,571,564	3,290,424	375,889	455,431	831,320	21,944	14,472	36,416
West Bengal	341,403	362,907	704,310	2,155,337	2,318,585	4,473,922	140,777	445,026	585,804	348,312	271,794	620,107	2186	1122	3308

Table 5 Estimated DALYs distribution by gender and major group causes in India and selected major states, 2020

Disease Categories	INMPDs			NCDs			Injuries			Ill-defined other			Covid-19 and other new		
Gender	Male	Female	Person	Male	Female	Person	Male	Female	Person	Male	Female	Person	Male	Female	Person
India	32,020,901	31,343,223	63,364,125	89,520,436	88,199,838	177,720,273	15,499,362	15,021,685	30,521,047	12,201,497	10,581,238	22,782,735	7,775,667	4,511,113	12,286,780
Andhra Pradesh	927,332	750,033	1,677,365	4,409,194	3,869,097	8,278,292	310,910	245,514	556,424	80,351	74,530	154,881	444,146	273,213	717,359
Assam	849,401	1,004,152	1,853,553	2,562,111	1,439,766	4,001,877	616,937	2,573,525	3,190,463	26,929	30,663	57,593	51,834	131,704	183,538
Chhattis-garh	687,053	790,341	1,477,395	2,021,115	2,357,381	4,378,495	229,602	283,369	512,972	842,862	853,040	1,695,902	160,163	100,239	260,402
Delhi	233,803	241,414	475,216	705,386	752,965	1,458,351	114,518	142,446	256,964	161,969	121,823	283,791	105,991	68,722	174,712
Gujarat	1,977,064	1,547,747	3,524,811	4,771,898	4,126,701	8,898,599	284,810	297,328	582,138	695,347	600,946	1,296,293	180,272	101,298	281,571
Haryana	491,208	391,103	882,310	1,842,641	1,526,764	3,369,405	384,722	220,117	604,839	1,123,191	705,288	1,828,480	77,415	48,738	126,153
Himachal Pradesh	137,820	88,637	226,458	526,952	497,707	1,024,659	57,293	49,090	106,383	97,651	75,164	172,814	82,035	47,285	129,319
Jharkhand	684,416	568,564	1,252,980	1,870,106	2,466,375	4,336,481	868,803	1,824,832	2,693,635	27,947	24,337	52,284	127,734	62,406	190,140
Karnataka	1,203,077	1,021,851	2,224,928	4,886,919	3,940,532	8,827,452	772,701	655,362	1,428,064	134,165	138,501	272,667	435,092	214,728	649,820
Kerala	250,758	151,291	402,048	2,908,948	2,763,543	5,672,490	345,715	384,710	730,425	194,627	118,038	312,665	183,467	127,586	311,053
Madhya Pradesh	3,458,409	3,100,250	6,558,659	4,854,882	4,605,172	9,460,054	658,077	876,238	1,534,315	2,914,995	2,602,115	5,517,111	161,448	79,143	240,591
Maharash-tra	1,505,760	1,426,306	2,932,066	7,148,675	6,664,603	13,813,278	1,505,634	408,826	1,914,460	754,184	490,524	1,244,709	1,359,992	743,130	2,103,122
Odisha	1,573,049	1,431,170	3,004,219	4,140,018	4,249,807	8,389,825	331,634	551,336	882,970	695,373	617,348	1,312,721	115,461	55,802	171,263
Punjab	758,561	623,418	1,381,979	2,182,348	1,718,709	3,901,057	300,061	314,057	614,118	174,686	97,640	272,327	291,113	134,914	426,027
Rajasthan	2,339,598	2,698,565	5,038,162	3,666,221	3,652,901	7,319,122	819,376	977,514	1,796,890	597,059	692,784	1,289,843	242,183	155,924	398,107
Tamil Nadu	853,749	643,071	1,496,820	5,451,150	4,447,541	9,898,691	524,294	500,570	1,024,864	559,012	441,497	1,000,508	143,343	56,501	199,844
Telangana	530,440	413,745	944,185	2,089,484	1,699,319	3,788,804	642,653	718,510	1,361,163	142,064	86,731	228,795	154,274	85,318	239,592
Uttara-khand	252,450	248,837	501,287	830,052	771,997	1,602,049	54,728	78,141	132,869	54,282	51,719	106,001	116,416	66,822	183,238
Uttar Pradesh	8,629,703	11,179,486	19,809,189	10,949,966	11,071,324	22,021,290	4,950,409	4,024,120	8,974,529	1,085,114	1,079,308	2,164,422	2,900,273	1,958,062	4,858,335
West Bengal	1,881,200	1,634,611	3,515,810	5,820,611	5,017,437	10,838,049	519,019	698,937	1,217,957	1,005,505	644,115	1,649,621	288,948	151,754	440,702

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Table 6 Estimated DALYs distribution by age-gender and major causes specific in India, 2020

Major Diseases Group	Gender	< 1	1–4	5–14	15–24	25–34	35–44	45–54	55–64	65–69	70 +	TOTAL	Share (%)
INMPDs	Male	17,851,025	1,172,667	980,675	1,441,280	1,760,590	1,915,080	1,966,335	2,043,907	969,076	1,920,266	32,020,901	20
	Female	16,637,074	1,162,832	1,207,064	2,388,969	2,040,705	1,958,749	1,802,711	1,725,818	756,281	1,663,022	31,343,223	21
	Person	34,488,099	2,335,499	2,187,738	3,830,249	3,801,294	3,873,829	3,769,046	3,769,725	1,725,357	3,583,288	63,364,125	21
NCDs	Male	3,656,734	2,571,584	5,028,650	7,256,451	8,711,238	11,166,323	12,866,414	14,133,843	6,941,139	17,188,059	89,520,436	57
	Female	3,112,356	2,568,381	5,088,354	8,007,464	10,133,019	9,768,140	10,250,064	13,156,186	6,705,496	19,410,377	88,199,838	59
	Person	6,769,090	5,139,965	10,117,005	15,263,915	18,844,257	20,934,463	23,116,478	27,290,030	13,646,635	36,598,436	177,720,273	58
Injuries	Male	300,689	410,531	463,669	2,380,543	2,521,778	2,150,241	1,907,060	1,720,671	660,856	2,983,325	15,499,362	10
	Female	232,598	858,310	1,496,450	3,933,088	3,267,423	1,564,258	960,582	975,795	313,972	1,419,210	15,021,685	10
	Person	533,287	1,268,841	1,960,119	6,313,631	5,789,200	3,714,499	2,867,642	2,696,466	974,828	4,402,535	30,521,047	10
Ill-defined other	Male	1,288,268	508,963	542,699	1,195,379	1,327,732	1,464,883	1,454,127	1,682,332	718,599	2,018,514	12,201,497	8
	Female	1,347,421	719,371	579,815	925,262	890,874	837,195	996,692	1,330,026	617,003	2,337,581	10,581,238	7
	Person	2,635,689	1,228,333	1,122,514	2,120,640	2,218,606	2,302,078	2,450,819	3,012,359	1,335,602	4,356,095	22,782,735	7
Covid-19 and other new	Male	61,207	38,931	92,623	285,922	468,829	779,985	1,217,110	1,792,644	932,750	2,105,665	7,775,667	5
	Female	34,903	40,099	95,184	197,019	289,042	382,271	645,600	1,075,841	528,168	1,222,986	4,511,113	3
	Person	96,111	79,029	187,807	482,941	757,871	1,162,256	1,862,710	2,868,484	1,460,919	3,328,651	12,286,780	4
All Causes	Male	23,157,924	4,702,675	7,108,317	12,559,575	14,790,167	17,476,511	19,411,047	21,373,397	10,222,421	26,215,828	157,017,863	100
	Female	21,364,352	5,348,992	8,466,866	15,451,802	16,621,062	14,510,613	14,655,648	18,263,666	8,920,920	26,053,176	149,657,097	100
	Person	44,522,276	10,051,667	15,575,183	28,011,377	31,411,229	31,987,124	34,066,696	39,637,063	19,143,341	52,269,004	306,674,960	100

Table 7 Age-wise leading causes of disease burden (DALYs) by group and individual causes in India, 2020

Rank	Group top Causes	< 1	1–5	5–14	15–24	25–34	35–44	45–54	55–64	65–69	70 +	All
1	Cardiovascular diseases-Circulatory system	1,250,646	689,758	898,723	2,010,332	2,947,954	4,849,693	7,072,165	9,925,603	5,514,091	15,457,075	50,616,040
2	Maternal and neonatal disorders	30,065,291	4679	0	715,801	592,688	387,638	421,420	0	0	0	32,187,517
3	Chronic respiratory diseases	2,245,546	1,699,596	4,659,280	2,438,605	2,587,772	2,360,845	2,869,561	3,587,240	1,771,388	4,668,193	28,888,026
4	Infectious diseases	2,389,109	1,039,474	983,533	1,482,094	1,719,596	1,936,022	2,006,903	2,045,252	936,386	1,990,593	16,528,962
5	Neurological disorders	530,306	808,521	1,907,315	1,964,599	2,296,008	2,639,475	2,002,077	1,201,902	499,635	770,297	14,620,135
6	Cancer and other Neoplasms	72,221	180,444	261,530	323,631	477,961	816,904	1,265,594	1,693,955	746,383	1,598,396	7,437,018
7	Digestive diseases	229,253	187,267	225,322	460,873	921,604	1,329,233	1,372,844	980,959	344,693	699,790	6,751,838
Rank	Individual top causes	< 1	1–5	5–14	15–24	25–34	35–44	45–54	55–64	65–69	70 +	All
1	Diabetes mellitus	21,323	30,643	66,712	207,071	744,082	1,513,261	2,566,330	3,694,422	1,779,706	4,122,820	14,746,370
2	COVID-19, Virus Identified	83,910	42,324	64,464	221,329	458,752	930,760	1,632,786	2,614,544	1,371,338	3,164,937	10,585,145
3	Ischemic heart disease	45,329	27,075	32,166	139,854	353,134	813,535	1,410,436	2,200,817	1,246,631	4,211,343	10,480,321
4	Hypertensive heart disease	26,001	14,824	28,499	96,492	203,624	446,138	844,326	1,085,623	1,015,941	1,888,925	5,650,393
5	Tuberculosis	66,884	81,954	201,165	543,406	551,021	548,402	458,385	397,562	172,233	278,924	3,299,937

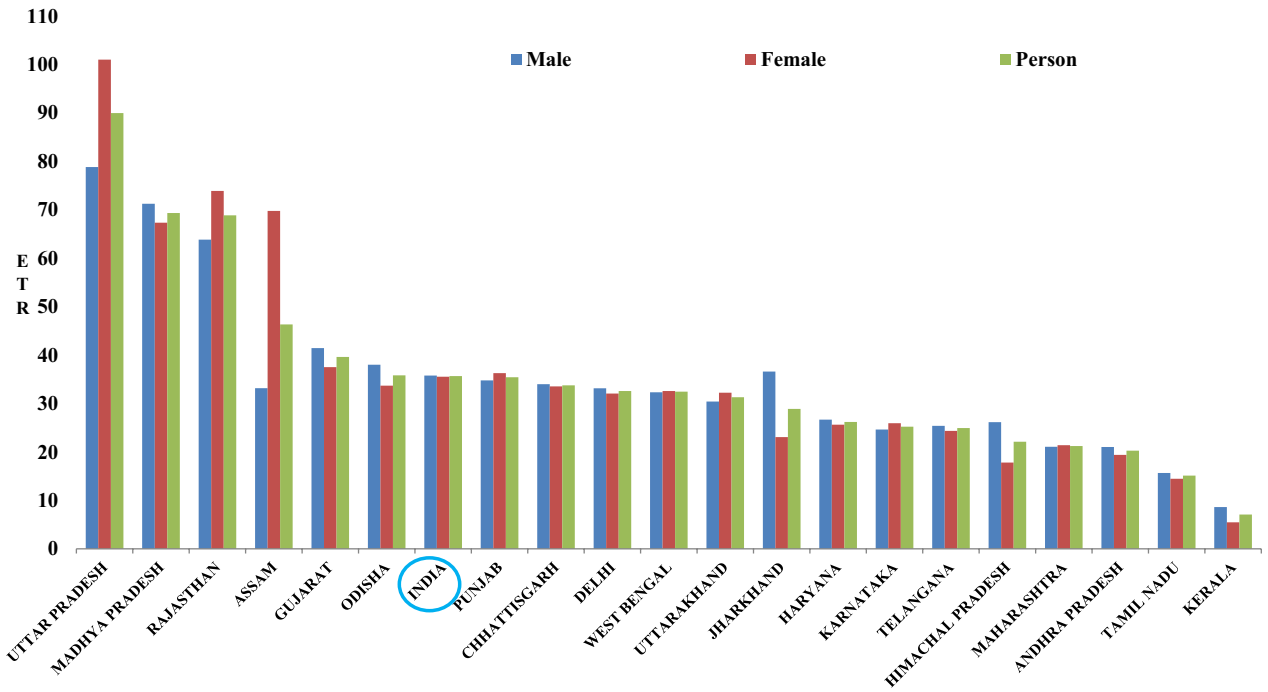


Fig. 1 Epidemiological transition status through ETR by gender at the national and sub-national level in contemporary India, 2020

at 46 (male: 33 and female: 70). Conversely, the lowest ETR is found in Kerala at 7 (male: 9 and female: 5), indicating a lower relative burden of INMPDs compared to much higher burden of NCDs. Other states with low ETR values include Tamil Nadu, Andhra Pradesh, Maharashtra, Himachal Pradesh, Telangana, and others, reflecting a more advanced stage of epidemiological transition with a predominant burden of NCDs.

Figure 2 is similar to Fig. 1 but focuses on children under five. It shows significant variation in ETR values across Indian states, ranging from almost 50 to 900, indicating a much higher burden of INMPDs than NCDs in this age group. Among selected states, Odisha has the highest ETR value at 788 (male: 897 and female: 690), followed by Madhya Pradesh at

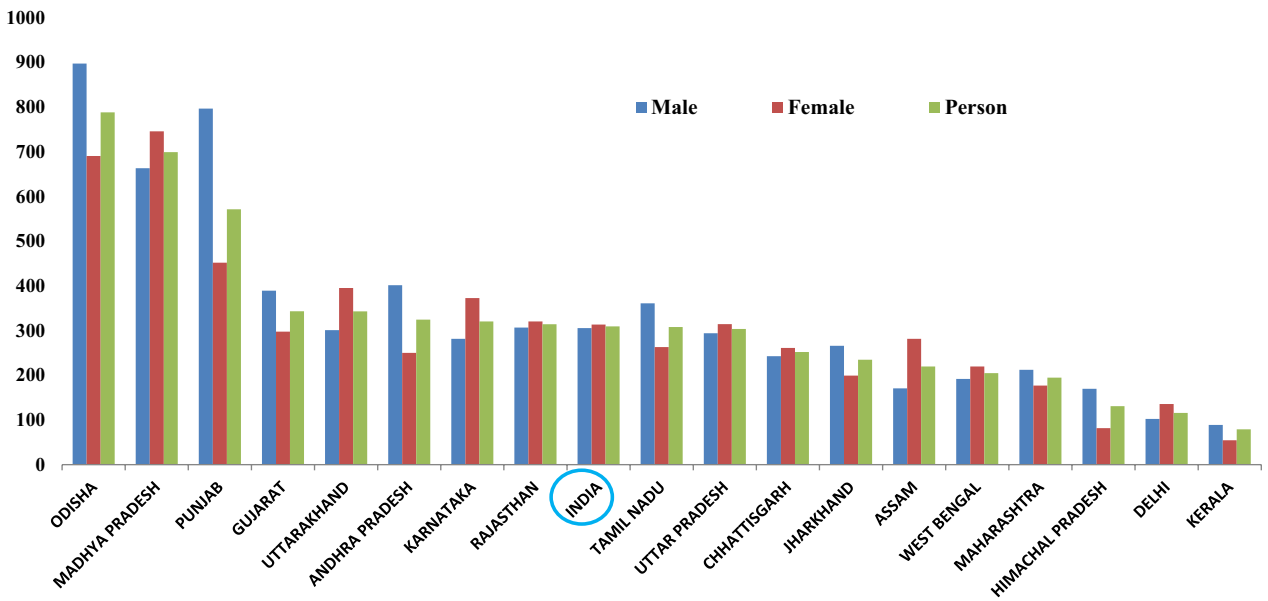


Fig. 2 Epidemiological transition status through ETR by gender for under-five children at the national and sub-national level in contemporary India, 2020

700, Punjab at 571, Gujarat at 343, and other states. Conversely, Kerala consistently has the lowest ETR value at 79 (male: 89 and female: 55). Nationally, the ETR value is 309 (male: 305 and female: 313). Similarly, Fig. 3 represents the national level ETR for individuals aged five and above. It demonstrates a substantial decline in ETR values with increasing age, dropping from 30 to 10 in India, which indicates an increasing burden of NCDs with age. Across ages, females have higher ETR values until the 45–54 age group, after which males have higher ETR values.

4 Discussion

The present study employs a novel approach to estimating indirect mortality measures, i.e., YLLs, YLDs, DALYs and Epidemiological Transition Ratio (ETR) at national and state levels in India, using limited data and existing studies. The study's estimates and patterns were validated through findings from existing literature [3, 12]. In a country like India, where minimal valid data on mortality and causes of death were available, especially at the sub-national level. Some studies based on the Million Death Study (MDS) data exist but are inaccessible in the public domain [25]. Other studies, such as the GBD, are complex, multi-country, multi-data, and several assumption-based. The current study's approach was to employ and overcome data limitations effectively. Furthermore, the methodology has been validated and endorsed by the ICMR Disease Burden Initiative [12]. However, we have modified this method based on data availability in the public domain, keeping the rationale behind the same. Additionally, our findings closely align with previously estimated measures, though derived for different objectives, further reinforcing the reliability and robustness of our approach. This method offers valuable insights into health status and disease burden, particularly in regions with limited data availability, making it especially useful for lower-middle-income countries.

The present study revealed an alarming situation in India, where NCDs accounted for nearly six out of every ten deaths in 2020. Another study observed a similar pattern half a decade ago but was lower than the current situation, indicating the increasing burden of NCDs over time [26]. India presents a "Nations within a Nation" setting due to its multiple states with heterogeneous populations across different states. The country faces a significant burden of NCDs; however, INMPDs remain dominant in many areas. Injuries continue to be a major concern in the middle working ages, and there is an ongoing threat from emerging diseases such as COVID-19 diseases [3, 12, 26]. Our estimates showed that the most comparable and valid measure of burden, DALYs, was around 306 million in India. Of these, NCDs account for the highest share at approximately 58%, followed by INMPDs at 21%. Other Injuries, Ill-Defined conditions, and COVID-19 contribute between five to ten percent of it. It infers that India is coping with a high level of epidemiological transition, with chronic illnesses comprising more than half of the disease burden and the emergence and remerged of infectious diseases [27–29]. Existing and current study noted the transition from communicable diseases observed at the state level

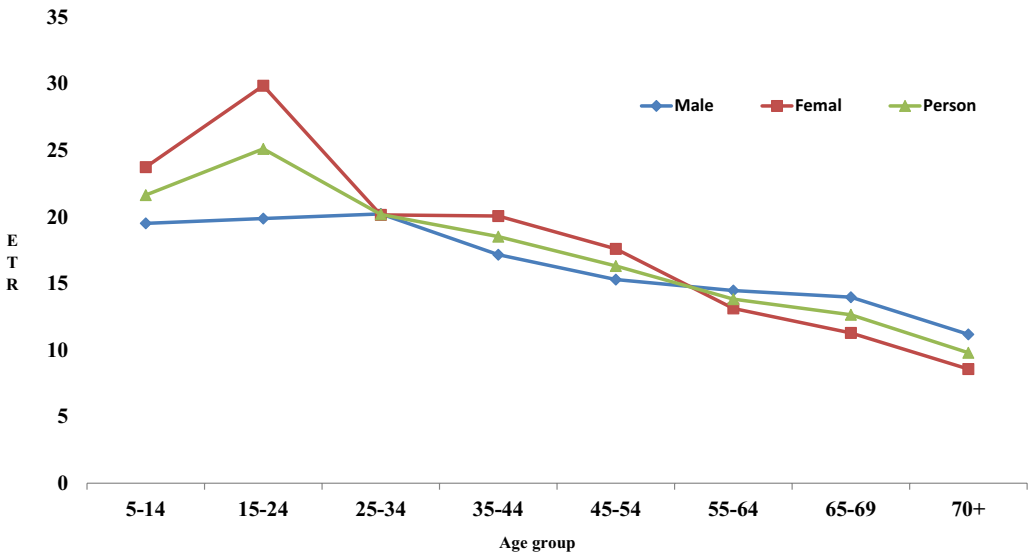


Fig. 3 Epidemiological transition status through ETR by age and gender in contemporary India, 2020

to chronic diseases [10, 30, 31]. This shift underscores the growing need to address chronic diseases in public health strategies as the dominant scenario of morbidity expansion [32].

Moreover, this study discussed that age and sex-specific all-cause mortality, YLLs, YLDs, and DALYs were higher at the early newborn ages of life (infant age). These metrics show a decline with increasing age but begin to rise sharply from 45 years onward, reaching their highest levels among individuals aged 70 and above in India.. This age-specific mortality pattern was linked to a significant proportion of infectious, contagious, neonatal, perinatal, nutritional, and maternal causes in early ages. These show a similar death pattern, as discussed by another study [33]. NCDs, such as cardiovascular diseases, diabetes, respiratory illnesses, and other conditions, are dominantly onset in adults aged 45 and above [12, 34, 35] in India. This study also highlighted key factors related to specific causes of death, YLLs, YLDs, and DALYs, emphasising sex differences in mortality and disease patterns. Males experienced higher deaths, YLLs, and DALYs across almost all age groups, while females had higher YLDs. It revealed that men are more prone to premature mortality due to the key cause of NCDs and Injuries. In comparison, women are more prone to living more years with chronic illnesses and disabilities, maybe due to their higher life expectancy compared to men in India and morbidity expansion. On the same line, supported findings were discussed in the existing literature in different contexts [30, 35–37].

This study has also enlightened on group and individual top causes of death in India, where CVDs are the topmost cause of disease burden. CVDs are a consistent burden grappling with circulatory diseases, ischaemic heart disease, cerebrovascular such as stroke, and heart disease-related problems prominent prevalent and fatal causes among the Indian population. It has consistently been the top cause for the past decade and will continue to be the leading cause in the future, as quantified by our and other studies [38, 39]. Similarly, we found that maternal and neonatal disorders were the second most common cause of disease burden. It found some inconsistencies with other literature, which showed that neonatal and maternal mortality declined over the period [40, 41]. However, our study provides a burden on the absolute DALYs that demonstrate the actual burden. Other leading causes are Chronic Respiratory Diseases, Infectious diseases, Neurological disorders, Cancer and Neoplasms. They are significantly prevalent in the spare of India's disease burden. At present, cancer is very chronic, fatal and expensive to treat. It is the sixth-highest cause of death in India; on the other hand, the growing aged population creates huge future concerns. As per a cancer ICMR report, one in nine people is expected to face a cancer diagnosis during their lifetime in India [42]. Overall, India is still not overcoming the dual burden of diseases escalating the burden of NCDs without much replacing the maternal, neonatal and Infectious problems [27, 43]. Likewise, this study identifies the top five causes of disease burden in India as Diabetes Mellitus, COVID-19, Ischemic Heart Disease, Hypertensive Heart Disease, and Tuberculosis. Notably, COVID-19 was the second leading cause in 2020, primarily due to loss of life, as highlighted by another study [44, 45]. During the study period, COVID-19 also considerably impacted the share of existing diseases, especially CODs due to NCDs. Except for COVID-19, other causes have been the most suffered diseases in the Indian population, more prominently persons aged 45 and above [3, 30].

In recent years, several state-level disease burden collaborative studies have been conducted in India to address various data gaps and achieve the specific aims of each study [3, 12, 46]. These efforts are crucial for providing more accurate and comprehensive insights into the health challenges faced by different regions, ultimately aiding in better planning and resource allocation. Significant heterogeneity in mortality and morbidity at the state level makes it essential to conduct research periodically to explore these gaps for better planning [15, 30, 47]. This study extends the analysis of disease burden at the state level from another perspective, revealing substantial variation across states. The burden of NCDs was most prominent, with states such as Uttar Pradesh, Maharashtra, Madhya Pradesh, Tamil Nadu, and West Bengal exhibiting higher burdens based on YLLs, YLDs, and DALYs for each disease. These states may exhibit higher burdens due to their large population share. Additionally, there are notable differences in the mortality patterns and DALYs among states.

The concept of ETR is validated through the framework of the epidemiological transition theory and rationally aligns with another study [3], which derived the epidemiological transition level. The present study found that India is at a higher-middle stage of epidemiological transition. It has a significant burden of NCDs rather than INMPDs shifting from acute to chronic illnesses [30, 48, 49]. At the country level, the ETR was not different from a gender perspective. However, there are considerable differences at the age-specific transition level. Similarly, the state-level epidemiological scenarios evaluate from low to high levels of epidemiological transition [3, 12]. From the selected states, Uttar Pradesh (UP) found the lowest epidemiological transition status, where the ETR value at the highest indicates that UP is still grappling with the dual burden of NCDs and INMPDs [50, 51]. Among Indian states, Uttar Pradesh is the most populous and lags in several development indicators. The main causes of burden in the state include COPD, diarrheal diseases, ischemic heart disease, tuberculosis, preterm birth complications, iron deficiency anaemia, etc., as previously reported. [52, 53].

Followed by UP, Madhya Pradesh, Rajasthan, and Assam, several other states are at a lower epidemiological stage. In contrast, Kerala is at the highest epidemiological stage with its advanced demographic and development indicators [54,

55]. Here, the major health burden has shifted towards NCDs [30], while problems related to communicable diseases, maternal health, and child health are minimal. Underdeveloped regions facing health challenges with infectious diseases, malnutrition, and maternal and child health issues can benefit from adopting Kerala's model or strategies from developed countries for improved health transitions. [56, 57]. This approach could help them progress towards better health for all.

5 Conclusion

The present study adopts a unique approach to evaluating disease burden using limited existing data, offering valuable insights for lower-middle-income countries lacking such information. This analysis highlights the significant variation in the burden of diseases across India and selected states, assessed through various dimensions in the contemporary period. Nationally, NCDs account for the highest proportion of deaths, followed by INMPDs, Injuries, Covid-19, and other emerging diseases. NCDs accounted for nearly six out of every ten deaths. The study also concludes substantial differences in disease burden based on age and gender within the Indian population. The study concludes that women experience a higher disease burden from reproductive and maternal health issues and chronic conditions, while men are more affected by premature mortality due to injuries and NCDs. Notably, the disease burden is significantly higher during the early stages of life, particularly under one year, and escalates again in later years, underscoring the need for age-specific and gender-sensitive health interventions.

Furthermore, the epidemiological transition status varies across sub-national levels, reflecting the varying magnitudes of NCDs and INMPDs burden. The study revealed that India and its states underwent multifarious epidemiological transitions at various levels in the contemporary period. There is at least a decade-long gap in the epidemiological transition among Indian states, with states like Uttar Pradesh, Madhya Pradesh and others lagging and having dual disease-burned scenarios compared to more advanced states such as Kerala. It highlights significant regional disparities as the disease burden shifts markedly, necessitating tailored health strategies to address these evolving patterns effectively. This work can support the ongoing planning efforts of various state and central governments to address the current burden of morbidities.

6 Policy recommendations

Achieving SDG target 3.4 to reduce premature mortality from NCDs requires adopting successful strategies from developed nations and strengthening India's health system through preventive and targeted measures. By looking at India's disease burden and epidemiological transition status, key priorities include improving health infrastructure, enhancing care quality, increasing hospitals and health workers through public–private partnerships, and extending medical facilities to rural areas. Conducting innovative research and developing robust strategies are crucial to managing NCDs and emerging diseases in India's heterogeneous and ageing population. Strengthening the National Programme for Prevention and Control of NCDs (NP-NCD) alongside the National Health Mission is essential to address urban and rural health challenges.

While data reporting in India has improved over time, it remains minimal, necessitating further efforts to enhance the accuracy of mortality and cause-of-death data, specifically at the sub-regional level. A stringent monitoring and evaluation system is required to ensure better planning, effective interventions, and high-quality research supporting the future health burden. There should be a targeted approach; most Indian states are at a middle or higher epidemiological stage, while some face a dual burden of NCDs and communicable diseases at a lower stage.

The present study revealed at least a decade-long disparity in epidemiological transition levels between higher and lower ETRs at the state level in India. Therefore, health policies and strategies must be customised to address each state's disease burdens and transition level. Some states lagging in the epidemiological transition may adopt the Kerala model or the other developed Western world for various socio-demographic developments. It can be a better example of the roadmap towards advance transition. There is a need to strengthen existing programmes and policies that implement nationwide awareness and prevention programs with targeted approaches at the sub-national level. Also, as India is the most populous country, there is a need to establish accessible and affordable early disease periodical screening up to the Primary Health Centre (PHC).

Further, there is a need for well-established monitoring and implementation of existing health insurance under state-centre partnerships, support for extensive research and innovation, and formulation of state-specific health policies.

Given the current scenario, there is a pressing need to strengthen health systems and encourage individuals and communities to adopt and prioritise healthier lifestyles and food. Since COVID-19, India has made significant improvements in its health system. However, the focus remains primarily on screening and treatment management for CVDs, cancer, and other chronic illnesses. Similarly, several high-ETR states still require substantial enhancement on the objectives of National Health Mission (NHM) services to address issues related to pregnancy complications, complete institutional delivery, reproductive health, infectious diseases, maternal care, and perinatal conditions. In addition to disease control-oriented policies, there is a growing emphasis on addressing injuries (both accidental and non-accidental), which remain a significant concern. These injuries resulted in India losing its most productive years, particularly among the middle-aged population. By focusing on these areas, policymakers can improve health outcomes, reduce premature mortality from NCDs and other causes, and effectively manage the dual burden of diseases in an ageing India. It may help to tackle the rising burden of NCDs in an ageing population without neglecting communicable diseases and other problems.

7 Limitation

The present study has quite a few limitations due to mainly data constraints. First, during the estimation of measures such as population, death rates, YLL, YLD, and DALYs, the analysis relied on demographic and public health assumptions that may affect the accuracy of the estimates. Further, proxy measures and indirect techniques were used for some indicators due to the limited data availability on specific indicators for particular periods. In our study, all the estimates were provided as point estimates only. We could not provide their standard errors or confidence intervals because the macro estimates were based on restricted information. In India, there is still an issue with the incompleteness of vital statistics. All analyses concerning the causes of death proportion were based on data from the MCCD, the only publicly available source in India. However, state-level reporting on causes of death is limited. Lastly, the calculation of YLD utilised indirect methods based on the ratio of YLD to YLL, derived from recent studies mapping the same diseases, which may introduce further variability in the results.

In comparison, our methodology is validated and aligns closely with previous measures. It involves indirect estimation techniques, which could introduce assumptions that might affect the precision of specific estimates, especially at the subnational level. These limitations may create gaps for further research.

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Author contributions Authors have read and agreed to the manuscript. MB has main role in study design, carried out the statistical analysis, interpretation and drafting the manuscript. MD has key role in study design, review and drafting the manuscript.

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Data availability Data used in the study was secondary macro data which has been directly retrieved from Registrar General & Census Commissioner of India. Every year RGI and Civil Registration System (CRS) published the reports of Medical Certification of Cause of Death (MCCD) and Vital Statistics which has been publically available on their official site (https://censusindia.gov.in/census.website/).

Declarations

Ethics approval and consent to participate Not applicable, this is a secondary macro data based study. For this study no ethical approval was required to undertake this study.

Consent for the publication Not applicable for this study.

Competing interests All authors have no competing interests.

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